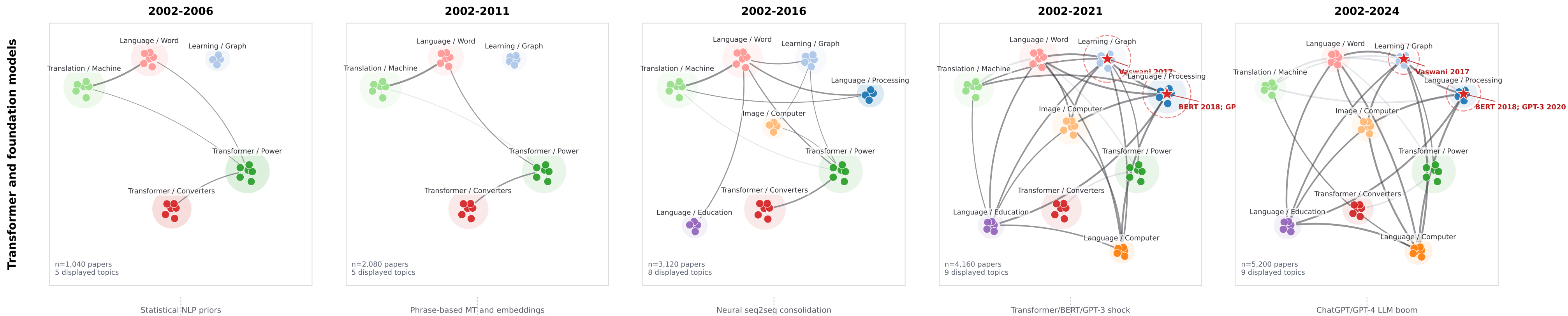


Fig. 1 | Citation knowledge-graph perturbation after a landmark innovation

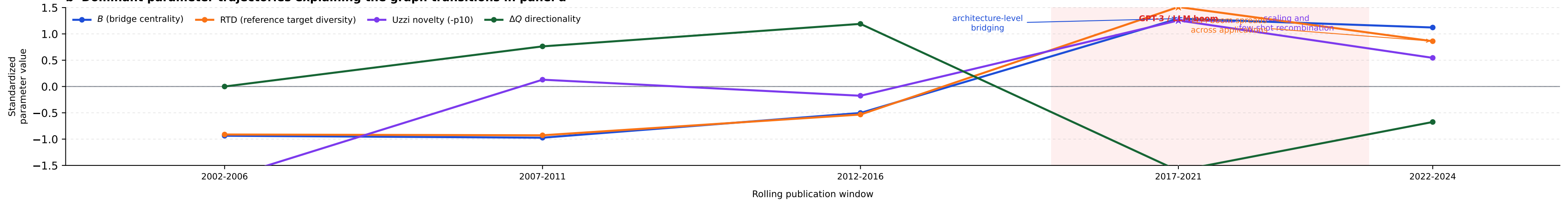
Example domain: Transformer and foundation models. Real OpenAlex data; display highlights data-derived topic snapshots and post-GPT-3 expansion.



a Cumulative citation knowledge graph snapshots



b Dominant parameter trajectories explaining the graph transitions in panel a



Bridging
B up + RTD up
Architecture and pretraining communities connect

Novelty
Uzzi up
Scaling and prompting recombine references

Expansion
edge gain + topic birth
LLM applications add visible mass